INCH POUND

MIL-R-11723E
20 October 1988
SUPERSEDING
MIL-R-11723D
13 July 1979

MILITARY SPECIFICATION

RADIATORS: HEATING, STEEL, MULTIFIN TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers multifin-type steel radiators, with mounting brackets and covers.

1.2 Classification. Radiators will be of the following sizes and lengths, as specified (see 6.2):

Size 1 - 1-1/4-inch nominal pipe size.
Size 2 - 2-inch nominal pipe size.
Length - 2-feet to 10-feet in 6-inch increments.
Length - 11-feet.
Length - 12-feet.

1.2.1 Part or identifying number (PIN). The specification number, size, and length are combined to form the specification PINs for the pump and radiators covered by this specification (see 6.5). The PINs for the radiators are established as follows:

	M11723	X	XXX
Specification number	↓		
Size (see 6.5)		↓	
Length (see 6.5)			↓

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer (Code 156), Naval Construction Battalion Center, Port Hueneme, CA 93043-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4520

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation-Packaging, Methods of.
MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-794 - Part and Equipment, Procedures for Packaging and Packing of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.2 Non-Government publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents which is current on the date of the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)

ANSI B1.20.1 - Pipe Threads, General Purpose (inch).

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A 120 - Pipe, Steel, Black and Hot-Dipped Zinc-Coated (Galvanized) Welded and Seamless, for Ordinary Uses.
ASTM A 178 - Electric-Resistance-Welded Carbon Steel and Carbon Manganese Steel Boiler Tubes.

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ASTM A 192 - Seamless Carbon Steel Boiler Tubes. For High-Pressure Service.
ASTM A 214 - Electric-Resistance-Welded Carbon Steel Heat-Exchanger and Condenser Tubes.
ASTM A 569 - Steel, Carbon (0.15 Maximum, Percent), Hot-Rolled Sheet and Strip, Commercial Quality.
ASTM D 3951 - Commercial Packaging. (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

HYDRONICS INSTITUTE

IBR Testing and Rating Code for Finned Tube (Commercial) Radiation.

(Application for copies should be addressed to the Hydronics Institute, 35 Russo Place, Berkeley Heights, NJ 07922.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification takes precedence. Nothing in this specification, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The radiator shall consist of a steel core equipped with steel fins. Unless otherwise specified (see 6.2), the radiator shall be furnished with a protective cover, mounting brackets, and fasteners.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2.1 and 6.2).

3.3 Standard commercial product. The radiator shall, as a minimum, be in accordance with the requirements of this specification and shall be the manufacturer's standard commercial product. Additional or better features which are not specifically prohibited by this specification but which are a part of the manufacturer's standard commercial product, shall be included in the radiator being furnished. A standard commercial product is a product which has been sold or is being currently offered for sale on the commercial market through advertisements or manufacturer's catalogs, or brochures, and represents the latest production model.

3.3.1 System of measurement. The dimensions used in this specification are not intended to preclude the use of the metric system of measurement in the fabrication and production of the material, individual parts, and the finished product, provided form, fit, and function requirements are satisfied.

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3.4 Materials. Materials used shall be free from defects which would adversely affect the performance or maintainability of individual components or of the overall assembly. Materials not specified herein shall be of the same quality used for the intended purpose in commercial practice. Unless otherwise specified herein, all equipment, material, and articles incorporated in the work covered by this specification are to be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products are allowed under this specification unless otherwise specified.

3.5 Design. The radiator shall be designed to permit easy accessibility for maintenance and service in the field. The design shall be such as to prevent condition hazardous to personnel or deleterious to equipment.

3.5.1 Capacity. The minimum heating capacity of the radiators shall be as shown in table I. The radiators shall be rated in accordance with the IBR Testing and Rating Code for Finned Tube (Commercial) Radiation.

TABLE I. Capacity - Steam rating Btu/hr/ft of finned length.

Bare radiators (painted but without cover)	
Radiator Core Size	Steam Rating at 214°F -- 217°F (BTUH)
1	990
2	1330

3.5.2 Operating conditions. The painted radiator with cover shall not be damaged when subjected to a working pressure of 0 pound-force per square inch gage (psig) to 150 psig with alternate heating and cooling from temperatures of 358 degrees Fahrenheit (°F) to between 60°F and 90°F.

3.5.3 Pressure. All elements of the radiator shall withstand 225 psig air pressure under water for 30 minutes without leakage or loosening of the fins.

3.6 Construction. The enclosure shall enclose the entire finned length of the radiator. The enclosure shall be fabricated from sheet steel of not less than 0.0478 inch thick and shall have openings that are not larger than 0.75 square inch in area. The enclosure shall not reduce the heating capacity of the radiator by more than 5 percent of the values specified herein and shall not warp when the radiator is subjected to the operating conditions specified herein. The enclosure may be in sections or as one piece. When sectional enclosure is furnished, the top and front sections shall be equipped with a fastening device.

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3.6.1 Core. The core shall be fabricated from the specified size (see 6.2), steel pipe or steel tubing. Pipe shall be standard weight (Schedule 40), black conforming to ASTM A 120. Tubing shall conform to ASTM Designation A 178, A 192, or A 214. For pipe, the size 1 cores shall have an outside diameter (od) of 1.625 inches with a wall thickness of 0.125 inch and the size 2 cores shall have an od of 2.375 inches with a wall thickness of 0.154 inch. For tubing, the core od shall be the same as pipe core od and the wall thickness shall be a minimum of 0.125 inches. At each end, the core shall extend not less than 2-1/2 inches beyond the fins. The core ends shall be either threaded or beveled for welding as specified (see 6.2). Threads shall be right-hand, external, taper pipe threads in accordance with ANSI B1.20.1. Both ends of the core shall be protected with metal or heavy plastic caps.

3.6.2 Fins. Unless otherwise specified (see 6.2), the steel fins shall be rectangular. The length or width of each rectangular fin shall be not more than 5-1/4 inches. When spiral steel fins are specified, the overall diameter shall be not more than 4-1/2 inches. The fins shall be secured to the core and shall not loosen, warp, or buckle when the radiator is subjected to the operating conditions specified herein. The fins shall be fabricated of steel conforming to ASTM A 569.

3.7 Treatment and painting. Unless otherwise specified (see 6.2), the radiator cover shall be painted and the brackets shall be treated or painted or treated and painted in accordance with the manufacturer's standard practice. The threaded surfaces and inside of the radiator core shall not be painted.

3.8 Workmanship. All parts of the radiators, including stampings, machined surfaces, and welded parts shall be clean and free from dirt, scale, flux, and other harmful extraneous material. The covers shall be free from rough or cutting edges or sharp objects.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to

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requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2.1).
b. Quality conformance inspection (see 4.2.2).

4.2.1 First article inspection. The first article inspection shall be performed on one radiator unit when a first article is required (see 3.2 and 6.4). This inspection shall include the examination of 4.4 and the tests of 4.5. The first article may be either a first production item or a standard production item from the supplier's current inventory provided the item meets the requirements of the specification and is representative of the design, construction, and manufacturing technique applicable to the remaining items to be furnished under the contract.

4.2.2 Quality conformance inspection. The quality conformance inspection shall include the examination of 4.4, the tests of 4.5.1, and the packaging inspection of 4.6. This inspection shall be performed on the samples selected in accordance with 4.3.

4.3 Sampling. Sampling and inspection procedures shall be in accordance with MIL-STD-105. The unit of product shall be radiators. All radiators offered for delivery at one time shall be considered a lot for the purpose of inspection. If an inspection lot is rejected, the contractor may rework it to correct the defects, or screen out the defective units, and resubmit for a complete reinspection. Resubmitted lots shall be reinspected using tightened inspection. If the rejected lot was screened, reinspection shall be limited to the defect causing rejection. If the lot was reprocessed, reinspection shall be performed for all defects. Rejected lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.3.1 Sampling for examination. Examination shall be based on inspection level inspection of S-2 and an Acceptable Quality Level (AQL) of 4.0 percent defective for major defects and 6.5 percent defective for minor defects.

4.3.2 Sampling for tests. Tests shall be based on inspection level S-2 and an AQL of 1.0 percent defective.

4.4 Examination. Each unit selected in accordance with 4.3.1 shall be examined to verify compliance with the requirements of this specification. Examination shall be conducted as specified in table II. Any radiator in the sample containing one or more defects shall be rejected, and if the number of defective units in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. All defective units shall be replaced with non-defective units prior to testing.

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TABLE II. Classification of defects.

Categories	Defects
Major:	
101	Parts missing or not as specified.
102	Component parts not interchangeable.
103	Construction precludes easy accessibility for maintenance or service.
104	Materials not of proper quality.
105	Dimensions not as specified.
106	Extended surfaces (fins) not properly bonded to tubes.
107	External connections not adequate.
108	Tubes not properly pitched to avoid trappings.
Minor:	
201	Painting of poor grade.
202	Workmanship not of high quality.
203	Identification marking not correct.

4.5 Tests. The first article, when a first article is required, and each sample selected in accordance with 4.3.2 shall be tested. Any unit failing to pass the following tests shall be rejected, and if the number of defective units in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Tests shall be conducted as outlined in the referenced documents as hereing specified.

4.5.1 Pressure test. Subject the radiator to an air pressure as specified in 3.6.3. Any evidence of leakage or separation of the fins from the core shall constitute failure of this test.

4.5.2 Capacity. Test the radiator for capacity in accordance with the IBR Testing and Rating Code for Finned Tube (Commercial) Radiation. Non-conformance to 3.5.1 shall constitute failure of this test.

4.6 Packaging inspection. The preservation, packaging, packing, and marking of the units shall be examined to determine compliance with the requirements of section 5 of this specification.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Methods of preservation. Cleaning processes, drying procedures, preservatives, and methods of preservation specified in the following paragraphs are listed in MIL-P-116 and shall conform to the requirements of MIL-P-116 and any applicable specifications.

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5.1.1.2 Cleaning and drying. Prior to the applications of preservative compounds or paint, surfaces shall be cleaned by process C-1 and dried by any applicable procedure of MIL-P-116.

5.1.1.3 Unprotected surfaces. Unpainted ferrous metal surfaces, including threaded surfaces, shall be coated with type P-21 preservative in accordance with MIL-P-116. Care should be taken not to coat fins, coils, or other areas that would impair the heat transfer capabilities of the equipment.

5.1.1.4 Accessories. Accessories shall be preserved and packaged method I or III as applicable in accordance with MIL-P-116.

5.1.1.5 Heating element. The heating element supply and return connections shall be sealed with type I, grade A, barrier material conforming to MIL-B-121 and secured with pressure-sensitive tape.

5.1.2 Commercial. The heaters shall be preserved and packaged in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or commercial as specified (see 6.2).

5.2.1 Levels A and B. Packing shall be in accordance with MIL-STD-794. Containers shall be selected from table I for the appropriate level.

5.2.2 Commercial. The heaters shall be packed in accordance with ASTM D 3951.

5.3 Marking. Marking shall comply with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The radiators covered by this specification are intended for use with hot water and steam heating systems to heat air under natural circulations (thermal convections) conditions.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Sizes, lengths, and quantities of radiators required (see 1.2 and 3.6.1).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. Military part numbers of the required radiators (see 1.2.1).
e. When cover, mounting brackets, fasteners are not required (see 3.1).

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f. When first article is required for inspection and approval (see 3.2, 4.2.1, and 6.4).
g. Whether threaded or beveled for welding core ends are required (see 3.6.1).
h. Whether other than rectangular fins are required (see 3.6.2).
i. Treatment and painting, if different (see 3.7).
j. Level of preservation and packing required (see 5.1 and 5.2).

6.3 Data requirements. When this specification is used in an acquisition and data are required to be delivered, the data requirements shall be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (CDRL), incorporated into the contract. When the provisions of DOD FAR Supplement, Part 27, Sub-Part 27.475-1 (DD Form 1423) are invoked and the DD Form 1423 is not used, the data should be delivered by the contractor in accordance with the contract or purchase order requirements.

6.3.1 Quality assurance acceptability. The following alternative method should be considered in lieu of the performance test as specified in section 4 when this specification is applied on a contract.

Reference paragraph:

4.5.2 - Alternative evidence of meeting the capacity requirement may be an IBR emblem and applicable data indicating compliance with the IBR rating program or a certified test report from a recognized independent test laboratory indicating the radiator has been tested and conforms to IBR Testing and Rating Code for Finned Tube (commercial) Radiation.

6.4 First article. When a first article inspection is required, the item will be tested and should be a first article sample or it may be a standard production item from the contractor's current inventory as specified in 4.2.1. The first article should consist of one unit. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, test, and approval of the first article.

6.5 Definitive specification PIN. The specification PIN is a definitive PIN which corresponds to the size and length of radiators covered by this specification and defines the requirements of the options presented under this specification. The specification number, and the size and length code number are combined to form the definitive specification PIN.

6.5.1 Cataloging data. For cataloging purposes, PINs for the radiators are assigned as follows:

	M11723	X	XXX
Specification PIN	———	|	|
Size code number	———	———	|
Length code number (see 6.5.2)	———	———	———

6.5.2 Size and length. The size and length of the radiator (see 1.2) are identified as one-digit and three-digit numbers (see table III).

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TABLE III. Code number to size and length.

Size 1	1	-
Size 2	2	-
Length	-	XXX

XXX is a three digit number corresponding to the radiator length in inches (see 1.2).

Examples: M11723-1-024 designates a size 1 (1-1/4-inch nominal pipe size) radiator 2 feet (24 inches) in length.
M11723-2-132 designates a size 2 (2-inch nominal pipe size) radiator 11 feet (132 inches) in length.

6.6 Subject term (key word) listing.

Steel
Mounting brackets
Covers

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodians:
Navy - YD
Air Force - 99

Review Activities:
DLA - CS
Air Force - 84

Preparing Activity:
Navy - YD

(Project 4520-0323)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-R-11723E	MILITARY SPECIFICATION - RADIATORS: HEATING, STEEL, MULTIFIN TYPE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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